

Instructional Objectives to Core-Gameplay: A Serious Game Design Technique

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ABSTRACT

This paper explores a gap within serious game design research. That gap is the ambiguity surrounding the process of aligning the instructional objectives of serious games with their core-gameplay i.e. the moment-to-moment activity that is the core of player interaction. We propose a core-gameplay focused design framework that can work alongside existing, more broadly focused serious games design frameworks. The framework utilises an inquiry-based approach that allows the serious game designer to use key questions as a means of clearly connecting instructional objectives to core-gameplay. The paper describes the use of this design framework in the context of a small section of gameplay from an educational game currently in development. This demonstration of the framework demonstrates how instructional objectives can be embedded into a serious games core-gameplay.

Author Keywords

Game Design; Gameplay; Instructional Objectives; Human Computer Interaction; Serious Games.

ACM Classification Keywords

K.8.0. Personal Computing: General – *games*. K.3.1. Computers and Education: General.

INTRODUCTION

Designing educational games can be a complex and challenging endeavor. The challenge comes from finding a balance between achieving player engagement and meeting learning outcomes [31]. Failure in finding this balance can result in a game becoming “chocolate-dipped broccoli” [5] i.e. forcing the integration of educational tasks with game design in the hope that it can be awkwardly molded into an engaging experience.

There are a number of educational game design frameworks that can assist with the design of educational games [2,6,13,30]. These frameworks offer various guidelines and structured processes to ensure developers make informed pedagogical decisions when creating educational games. The common theme with these frameworks is the importance of clearly defined learning objectives. Whilst these frameworks are useful in guiding the entire game design and development cycle, there exists a common gap with respect to the translation of high-level guidelines and criteria into practical game design elements. The gap appears with the design of gameplay – that is the challenges faced and actions performed to achieve the goals of a game [1]. More specifically the issue resides within the design of *core-gameplay* i.e. the moment-to-moment activity that is the core of player interaction. The core-gameplay should align with the learning content to ensure maximal learning and engagement [14]. However, there seems to be no clear guidance on how to systematically approach and produce good design decisions for the purposes of learning. There are some frameworks that have provided useful matrices designed to highlight similarities between learning outcomes and types of core-gameplay [3,11,13,26]. However, the connection between instructional objectives and core-gameplay design, i.e., how gameplay can be intentionally, systematically and methodically designed to meet the learning objective, remains ambiguous.

A design framework is required that can work alongside existing educational game design frameworks, and provide additional support in guiding designers to create core-gameplay that is based on clearly defined instructional objectives. Using this framework can clearly identify how and why educational games will meet the desired learning outcomes. The potential of educational games is in its ability to illustrate educational content and harness motivational characteristics of the player. The framework presented provides a method for systematically aligning core-gameplay.

This paper is a *design exploration* grounded in the field of game design, education and human-computer interaction. We outline techniques for designing core-gameplay and provide details on the process that links core-gameplay interactivity to specific instructional objectives.

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The resultant framework is presented in the context of a small section of gameplay from an educational game currently in development. The design method aims to bring clarity to the process of identifying and molding game activities that align with desired learning outcomes for the player. In turn, it is intended that this knowledge will contribute to the wider educational games industry in their desire to build effective and engaging serious games.

LEARNING FROM GAMEPLAY

Serious games are digital games that have been designed to allow users to learn or demonstrate skills, knowledge and/or behaviors that have value beyond the game itself [19]. One of the most-powerful characteristics that digital games can bring to learning is the ability to increase motivation to engage with learning content [23]. Designing games to support autonomy, competence, relatedness [9] and Flow [8] can help facilitate enjoyment, ongoing gameplay [23,28] and consequently result in meaningful learning outcomes. With a pedagogical perspective, Habgood and Ainsworth [14] recommend that harnessing the motivational pull of digital games for learning can be achieved by tightly coupling the learning content with the core mechanics i.e. the series of mechanisms that are designed to be repeatedly used during gameplay. Evidence of this approach has shown that learning content coupled with core mechanics can create more effective learning outcomes [14]. Serious games are a valuable tool within a larger program containing a range of instructional techniques and tools [15].

Serious games can be used to facilitate a number of different pedagogical approaches such as constructivism, constructionism, situated cognition, deduction and hypothesis testing, complex concepts, abstract thinking, and visual and spatial processing [12]. Serious games also facilitate a number of different types of learning outcomes. In a systematic literature review of the evidence on serious games, Connolly et al. [7] categorised the behavioral and learning outcomes of serious games as affective and motivational, behaviour change, knowledge acquisition /content understanding, motor skills, and perceptual and cognitive, physiological and social/soft skill outcomes. This paper primarily discusses the design of games that have the goal of behavioural change and knowledge acquisition /content understanding.

Transfer of knowledge is based on the relationship between a person's sensing and memory systems. Short-term memory receives instructional content via the senses, is then processed by working memory and is potentially stored in long-term memory [22]. The qualities and design of the instructional content will determine if that new knowledge will be assimilated with existing knowledge (meaningfully) or will only exist as a separate concept (rote) in long-term memory [4].

The benefit to knowledge being learned meaningfully is that it is stored in long-term memory for longer and has potential to be used in further learning or problem solving contexts i.e. used outside of the learning context in real life [22]. These meaningful qualities consist of conceptually clear learning material, existing relevant prior knowledge to facilitate assimilation, and internal motivation to learn [22].

INSTRUCTIONAL OBJECTIVES

There is a need to understand how people learn in meaningful ways and it is necessary to examine how to create appropriate instructional objectives to meet this aim. Instructional objectives are statements that clearly outline intended instructional outcomes [18] and designing clear instructional objectives is an essential step when creating effective educational experiences [20]. Instructional objectives should outline three specific qualities that describe the outcome of the learning experience: performance, condition, and criterion. Creating statements that include these qualities starts by answering a series of questions.

- What should the learner be able to do? (Performance)
E.g., *the student will be able to ride a bike*
- Under what conditions do you want the learner to be able to do it? (Conditions)
E.g., *whilst in slippery conditions* [the student will be able to ride a bike]
- How well must it be done? (Criterion)
E.g., [whilst in slippery conditions the student will be able to ride a bike] *for 5 minutes without falling over*

Based on these questions an instructional objective could be "*whilst in slippery conditions the learner must ride a bike for five minutes without falling over.*"

During the serious game design process many instructional objectives might be formulated. However, not all objectives may be incorporated into one game experience. The core-gameplay framework offers designers a way to identify, synthesise and embed key instructional objectives within core-gameplay i.e. identifying instructional objectives that can be transformed into interesting choices and engaging interaction sequences.

SERIOUS GAME DESIGN FRAMEWORKS

Within a serious games context, motivation to play may be determined by the way the game's core mechanic links to the educational purposes of a game [14]. There are a number of educational game design frameworks that can assist with designing educational games and gameplay with guidelines, mapping tools and tables.

Some design frameworks take a holistic approach, encompassing many different elements that aim to guide the design of the entire game from start to finish [2,6,13,30]. These game design frameworks offer various guidelines

and structured processes to ensure developers make informed pedagogical decisions when creating educational games.

Other gameplay frameworks provide useful matrices that highlight similarities between learning outcomes and types of gameplay as defined by genre [11,25,26]. The mapping of learning objectives to game elements during the design has been explored in research using pedagogical foundations such as Bloom's revised taxonomy [16] as a lens to see how different game elements can be viewed in regards to the cognitive processes that occur during gameplay [26]. A typical gap in all these frameworks is the justification and reasoning behind the mapping of desired learning objectives to the designed gameplay in order to enable the player to reach those objectives.

The Learning Mechanic to Game Mechanic (LM-GM) framework [3] aims to address this gap by mapping learning content elements to elements within game mechanics. Applying this framework to the analysis of educational games helps to understand the relationship between learning objectives and gameplay. However, ambiguity remains in understanding how to design *core gameplay*, defined as the moment-to-moment activity within the game, to meet learning objectives. With applied design in mind, further investigation and clarity is required to ensure the core gameplay is guided by the desired learning outcomes.

CORE-GAMEPLAY AND INTERACTION MODELS

Gameplay fundamentally consists of two components: the challenges that a player must face arriving at the objective of the game, and the action that the player is permitted to take to address these challenges [1]. There is a close relationship between the gameplay within the game and its core mechanics, with the core mechanics being the "data and algorithms that precisely define the games central rules and internal operations" [1].

Interpretations of the precise nature of game mechanics vary across the academic and game design literature [1,3,17,27]. Sicart [27] discussed different types of game mechanics such as core, primary and secondary mechanics. The core mechanics can often be the combination of various activities that are repeatedly used to progress towards a rewarding end state. The experience that emerges from repeatedly interacting with the core mechanics is viewed as core-gameplay.

Understanding the interaction loop that occurs during the moment-to-moment experiences within the game is required to design core-gameplay. A common framework for dissecting interaction into stages is Norman's Seven Stages of Action [21]. The cycle can be used to understand the stages that occur leading up to the execution of an action (feed forward) and the impact of the action (feedback) [21]. An important way to use stages of action

as design aids can be to base questions around each step in the interaction cycle e.g. (Goal) "What do I want to accomplish?", (Specify) "What can I do?", (Perceive) "What happened?" [21].

The answers to questions such as these can provide a useful insight into how to design activities to meet predefined principles and guidelines e.g. Norman's Seven Fundamental Design Principles of discoverability, feedback, conceptual model, affordances, signifiers, mapping and constraints [21]. It is important to note that within the Seven Stages of Action there may be numerous sequences of interactions that occur before the original goal is met. In the context of a game example, a player may start by heading towards their goal of saving the princess, which can be achieved through the gameplay of defeating enemies leaping gaps. However, it is not until a deep understanding of core-gameplay is explained (e.g., jumping on top of enemies to kill them) that the Seven Stages of Action become an invaluable tool to explain what the player needs to do, what they have done, and how the system should respond.

It is important to understand that more than one core-gameplay interaction may occur in parallel or sequentially. It is during these moments that higher order cognitive processes may be facilitated such as analysis or evaluation [16]. For example, identifying (pointing out) threats, may transition into evaluative interactions (triaging threats) by attending to them in order of severity.

Whilst the Seven Stages of Action can help to understand the moment-to-moment interaction, the aim is to develop a game design process. Merging an understanding of interaction and core-gameplay into a framework allows the design of both. The elements used within this framework are based around the same interaction cycle found in the Seven Stages of Action; however, the language used speaks more clearly to the design of core-gameplay.

The constructed framework (see Figure 1) focuses on designing the elements of goals, choice, action, rules, and feedback. These essential elements of gameplay, game mechanics and the interaction cycle can be manipulated and sequenced by designers to form core-gameplay.

The framework works on the basis that players are presented with a goal, which involves a choice to be acted upon in order to progress towards their objective. Based on the player's previous knowledge and skill, the player determines the correct action to take and performs it. That action is evaluated by the system's rules and feedback is then sent back to the player. The player then interprets the feedback and potentially learns to adjust any further decisions made when presented with similar choices. The design of these qualities will determine if the player can utilise this new knowledge beyond the game-based situation.

The strength of the analysed gameplay is its relationship to how desired learning objectives maybe be represented in the game. Similarities can be drawn between the qualities required creating a learning objective and the elements required when constructing gameplay, game mechanics and the interaction cycle. These similarities are discussed in the following section.

FRAMEWORK ELEMENTS

Goals:

A core-gameplay goal can be guided by the instructional objective developed i.e. we wish the player to interact with and demonstrate a competent level of knowledge or skill in the form of a task performance. This performance is displayed by interacting with the core-gameplay. If players do not currently have the knowledge or skill to perform the task at the expected competency level, then either they can be supported through instructional techniques separate to the game. Alternately, the game could be incrementally developed using feedback during the various gameplay cycles. For example, when first interacting with the core-gameplay, the player may need to be guided using supportive feedback to determine the mechanisms of the core-gameplay. This feedback is often seen as a tutorial that explicitly outlines how the player can perform actions in the game. Strategies to overcome challenges may also be presented during this time. Therefore, if the core-gameplay is a mirror of instructional objectives then the player will inadvertently learn the knowledge and skills required to overcome the games challenges. Through various progressive iterations of the core-gameplay, feedback and support can be reduced to allow the player's knowledge and skills to be tested to the point where they are matching the expected level of competency.

The framework builds on this understanding of goals and specifies a question to be asked during the design process:

G1: What performance should I be able to achieve? [18].

This question allows the game designer to clearly articulate the instructional objective within a particular game context. The answer to this question will outline what the learner should be able to do competently by the end of the game.

Choice:

Choice is an integral cognitive process that occurs during the lead up to performing a task. During the planning and specifying stages of Norman's Seven Stages of Action [21] it is important to identify what alternatives are available when deciding to perform a task. Salen & Zimmerman [24] also discuss the anatomy of a choice utilising five fundamental questions. Of these five, the question of "How is the possibility of choice conveyed?" fits within our choice component of the framework.

Whilst discussing the components that form the absolute difficulty of a challenge/task, Adams [1] identifies that the intrinsic skill is an essential component that when combined

with stress, creates a level of difficulty. Intrinsic skill is the "level of skill needed to surmount the challenge if you give the player an unlimited amount of time in which to do it" [1]. This level of skill can be seen as the amount of mental or physical effort required to perform a task. The amount of physical effort in digital games is relatively low, and the amount of cognitive effort is relatively high. This cognitive effort can be viewed as the difficulty in determining the most acceptable option amongst the different task states and associated options [29]. Therefore, it is important to choose an instructional objective that can offer the player an adequate number of interesting and meaningful choices that are deserving of their effort.

Based on this understanding of choice two questions are included in the framework:

CH1: What can I do and what are my alternatives? [21]

CH2: How is the possibility of choice conveyed? [24]

The answers to these questions will outline the choices and the way in which choices are presented to the player.

Action:

Within the Seven Stages of Action [21] the result of enacting a choice occurs during the "perform" stage. Norman [21] proposes the question of "How do I do it?" can aid in detailing exactly how the user will interact with the system. How the user will "do it" may be either an individual or series of actions. From the perspective of core-gameplay, this observation is the detection of player actions by the system after a choice has been presented. Adams [1] recommends that verbs be used to reflect what the player does. These actions may not always map exactly to how they might appear in a real situations. Therefore, it is important when choosing actions that they conform to usability principles such as those found in Norman's Seven Fundamental Design Principles [21] so that they will still remain intuitive and easy to use.

Based on this understanding of action a question to ask during the design process has been incorporated into the framework:

A1: What series of action (verbs) will indicate I am making a choice? [1,18,21]

The answer to this question should be a series of verbs/steps that outline the player's interaction with the games interface.

Rules:

The rules in the framework are used as an evaluative method to ascertain the performance level during the core-gameplay. The learning objectives will outline the various elements that make up the "criterion" i.e. level at which someone has to be at to be considered competent [18]. These elements consist of the speed, accuracy and quality of the performed task and relate very closely to the

elements that make up difficulty in games. Adams [1] identifies intrinsic skill and stress (time pressure) as qualities that determine the amount of effort required overcoming the challenge. It is also important to outline the conditions and constraints under which the learner must perform the task e.g. what are the things they can/can't use and what setting, or physical environment does the performance need to be in [18].

There will be other game mechanics that also have rules; however the primary focus is on the core-gameplay and the related core-mechanics/rule systems that go with it. It is important to ask questions that identify the characteristics that form the basis of the rules that govern if a learner is competent enough to overcome the designed gameplay.

Based on this understanding of rules five questions to consider during the design process have been included in the framework:

R1: (Speed) How quickly do I need to make my choice and perform the actions to achieve my desired goal? [18]

R2: (Accuracy) To what degree or level of precision do I have to demonstrate to achieve my desired goal? [18]

R3: (Quality) What standards/criteria do I have to meet whilst I perform the actions to achieve my desired goal? [18]

R4: (Conditions) In what conditions will my performance occur? [18]

R5: (Constraints) What can/cannot I use whilst making a choice performing the actions? [18]

The answers to these questions will outline the criterion, conditions and constraints that will set the standard level of competency. Whilst the core-gameplay is being used to build up knowledge and skill, it is expected that this criterion level will start lower and buildup as the player progresses until they reach the standard outlined in the instructional objective.

Feedback:

While Mager's [18] learning objective structure does not identify how to construct feedback, there is a critical role feedback has in both the interaction cycle and learning. It is through feedback, both internal and external to the game, that learning can occur i.e. corrective feedback on previous task performances or through contextual events triggered by the system. To design feedback based around the choices a player makes, Salen and Zimmerman [24] identified certain questions as part of their anatomy of choice. In the Seven Stages of Action, Norman [21] also discusses questions that can be asked when designing for the three stages that form feedback.

We use this understanding of feedback to formulate key feedback-related questions within the core-gameplay framework:

F1: What is the result / new state of the environment once the choice and actions are made? [24]

F2: How do I know I made the correct/incorrect choice? [21,24]

F3: How will this affect my choices in the future? [24].

The answers to these questions will outline the system events that occur after players have made their choice and how they are expected to respond to future choices. It is at this point that constructive feedback can be given, and learning can be re-enforced.

FRAMEWORK USE

The framework developed is comparable to a user interaction framework, similar to those established in human-computer interaction (e.g., the Seven Stages of Action [21]). The purpose of articulating a core-gameplay framework is to establish a process whereby key game design strategies can have an apparent impact on learning. The process can be applied in a systematic way. By questioning every design decision raised by the framework it is anticipated that critical design thinking and idea generation that is guided by the required instructional outcomes will be stimulated. This type of thinking, in turn, leads to designed educational games that align to the instructional objectives, where each design decision can be clearly justified. This process is not context specific and is aimed at being flexible enough to fit in with existing educational game design frameworks that support the development of any educational game.

The usable representation of the working framework is structured around an open inquiry process where learning objectives, interaction design and gameplay questions are used to drive the design. These questions are aimed at investigating the desired outcome of performing a task at a level of competency. The answers then form the building blocks of the goals, choices, actions, rules and feedback. The following section outlines how this design process can be used to map one instructional objective to core-gameplay.

Design Brief: General Fire Evacuation Safety

Office based workplace safety training was the educational focus of this design exercise. An existing workplace safety training implementation, in the form of an e-learning module, already existed. This instructional method delivered relevant information via audio and visual presentation. The module was a self-paced slideshow that concluded with a short multiple-choice quiz.

Numerous consultations with workplace safety management staff highlighted the limitations of the current e-learning system and some key topics that would be useful to explore in a game-based experience. A prevalent workplace safety issue is that of injuries resulting from slips, trips and falls. These accidents lead to personal injury

costs to the injured employee and a loss in productivity and liability costs to business.

Expert discussions, existing training material, and various related safety documents were analyzed. From this analysis, instructional objectives to be pursued in the game were developed. Guided by the framework, a design process was then followed that allowed the alignment of core-gameplay with the instructional objectives.

The Game Example

At the time of development, the game was titled “Catastrophe”. Catastrophe is a top down crisis management game where the player manipulates the environment to keep non-player characters (NPCs) safe in normal working situation (Figure 1). Players can interact with in-game objects that are represented as potential safety hazards. These hazards all relate to those resulting in slips, trips and falls e.g. boxes or rubbish left in walkways, spilt liquid and loose cords. The NPCs will primarily act on their own, traveling between tasks such as working at a computer to handling boxes, rubbish or using the kitchen.

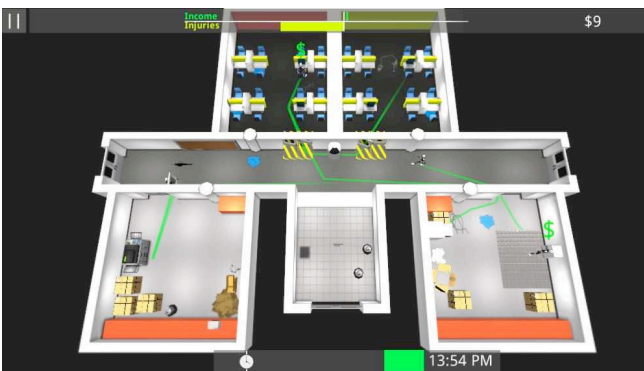


Figure 1: Catastrophe Crisis Management Game

The core-gameplay (moment-to-moment experience) requires the player to prevent injuries from occurring e.g. removing any hazards before an accident occurs. Game goals and concise safety information is delivered to the player via an advisor. If NPCs collide with a potential hazard, they will injure themselves. This will affect the player’s final game score through loss of productivity and injury costs. The game has been specially designed to be extendable if further learning goals are to be pursued e.g. evacuation planning, emergency communication (alarms), and the use of hazard fighting equipment (fire extinguishers).

A series of instructional objectives were designed based upon the following learning goal. “After the training, the learner will be able to identify various trips and falls within an office environment and know how to respond”. Whilst relatively open-ended, this learning goal gives an ideal starting point to develop a cognitive walkthrough that identifies the discrete steps and choices made when identifying and responding to hazards. From this map,

critical, yet interesting task decision points could be identified as the focus. These could then be developed into instructional objectives that clearly explain what the learner should be able to do after they play the game. An example of one of those instructional objectives will be the focus of the following example.

- What should the learner be able to do? (Performance)

Respond (applying control measures)

- Under what conditions do you want the learner to be able to do it? (Conditions)

Presented with a potential hazard in an office environment [the learner will respond (applying control measures)]

- How well must it be done? (Criterion)

[Presented with a potential hazard in an office environment the player will respond] choosing the ideal control measures to reduce the level of risk

The formed instructional objective might look like this: “When presented with a potential hazard in an office environment the learner will respond by choosing the ideal control measures that could reduce the level of risk.”

As part of the inquiry approach, the instructional objective is considered through a framework with questions that break down a task into its sub-components and qualities. The answers to these questions can then be used as building blocks to form the core-gameplay. Design challenges related to mapping learning to game elements becomes clear once the relationships between the instructional objective and core-gameplay are broken down.

The following tables outline how the elements of goals, choice, action, rules and feedback were constructed based around this specific instructional objective. The real life and game examples are focused on the desired level of competency i.e. the core-gameplay cycle experienced towards the end of the game.

Goal		
Question	Real Life	Game
G1: What performance should I be able to achieve? [18]	Respond to hazards by applying different control measures.	Respond to hazards by applying different control measures.

Table 1. Goal Mapping Table

G1: Within a real life context, the learner should be able respond to potential hazards. In an Occupational Health and Safety context, this response is known as applying control measures. The application of different control measures in the game allows the user to learn from the effect they have in a simulated environment and to develop a greater depth

of understanding. This application allows a tighter coupling of the decisions and actions experienced within the game so that they can may be transferred into a real situation.

Choice		
Question	Real Life	Game
CH1: What can I do, and what are my alternatives? [21]	Determine the ideal control measure/s to apply from the list of existing control measures.	At least 2-3 meaningful choices/control measures available to the player.
CH2: How is the possibility of choice conveyed [24]?	The perceived affordances of objects in the environment and the internally imagined actions taken with objects not currently present.	Menu of buttons with each available control measure present. Menu buttons will have tool tips to identify the future effects it will have.

Table 2. Choice Mapping Table

CH1: The focus of this cycle of core-gameplay is presenting the user with similar choices to those choices available when facing hazards in the real world. Realistically there will often be two-three control measures that could be applied. Therefore, the game offers the player the same types of choices that are core to responding to a hazard and ultimately helps to re-enforce making the most effective decisions. In the game, each hazard will have at least two or three choices that could have some effect on the hazard.

CH2: In real life, choices to interact with objects at hand are often conveyed by their perceived affordances [21], or some decisions could be made based on an understanding of objects that are not immediately available. For example, choosing to place a barrier around a hazard first requires an understanding that things nearby could be used as a barrier or that barriers are found elsewhere. The game simplifies the decision-making process by clearly conveying if the necessary actions and tools can or cannot be used. This design decision ensures the core-gameplay focuses on making the critical decision of choosing effective control measures. This choice will be visually conveyed via a menu of buttons that represent applying a control measure. Tooltips over each button will identify the future events that will occur once the button is pressed.

Action		
Question	Real Life	Game
A1: What series of action (verbs) will indicate I am making a choice? [1,18,21]	Series of steps that represent applying a particular control measures.	Click the hazard, then click the control measure button.

Table 3. Action Mapping Table

A1: The real life performance involves completing a sequence of tasks for each control measure that will achieve the goal e.g. cleaning up a spill requires fetching a sign to warn others, fetching cleaning products, applying cleaning products, and storing the sign and cleaning products when done. Direct mapping of these real actions would push the scope and focus of the game beyond the requirements. Therefore, the majority of actions taken for this core-gameplay cycle consist of clicking on the hazard, and then clicking on a control measure button.

Rules		
Question	Real Life	Game
R1: (Speed) How quickly do I need to make my choice and perform the actions to achieve my desired goal? [18]	Respond to a hazard as soon as it identified visually.	Respond to a hazard as soon as it identified visually.
R2: (Accuracy) To what degree or level of precision do I have to demonstrate to achieve my desired goal? [18]	One or more control measures will reduce the level of risk.	One or two control measures out of the six will significantly reduce the risk of a hazard injuring an NPC.
R3: (Quality) What standards/criteria do I have to meet whilst I perform the actions to achieve my desired goal? [18]	Apply the correct ordering of control measures based on an evaluation of the time a control measure might take to perform.	Evaluate what controls should have been chosen first based on each hazard. i.e. higher order controls may not always be the ideal first response.

R4: (Conditions) In what conditions will my performance occur? [18]	Real office environment, presented with a real hazards and workers.	Simulated 3D office environment presented with simulated hazards and workers.
R5: (Constraints) What can/cannot I use whilst making a choice performing the actions? [18]	Can't use reference material. Could ask for assistance if hazards cannot be immediately eliminated.	Can't use interactive safety reference material. Assistance could be provided when making and applying a choice.

Table 4. Rules Mapping Table

R1: The NPC AI routines will travel to office tasks that bypass hazards frequently. This design decision ensures that the player is motivated to respond to all hazards quickly before an NPC is injured. Some control measures will not immediately reduce the level of risk, re-enforcing the idea that temporary control measures may also be required.

R2: The accuracy criterion of the task is based on the predefined control measures that should be used for each hazard. For example, a hazard will have various control measures that can and cannot be applied. The game will always expect that at least one or two control measures are correctly chosen before the hazard no longer poses a risk of injuring the NPC.

R3: Players will be evaluated on the quality of their response based on their ability to determine the ideal control measure to apply first. Often lower order short term controls need to be put in place until higher order controls can be effectively applied. Some of the control measures will be more effective at reducing the risk of injury than others. For example, cleaning up a spill takes time but is more effective than placing a barrier around it. Ideally the barrier should be placed first to reduce the level of risk immediately.

R4: An environmental condition that could present the player with a multitude of hazards to respond to was used i.e. a 3D simulated office environment with animated workers, hazards and responses. The camera perspective used was a top-down view that offered the player a greater sense of choice to perform various actions as well as giving the player the ability to spot hazards immediately. It was also important to include NPC moving about the environment as this emphasized the importance or responding to hazards quickly and effectively.

R5: When performing the task at a level of competency in real life it is expected that the learner would not have any

physical reference material them to aid in determining the ideal control measures to implement. In some cases, not all hazards can be eliminated immediately. Assistance is often necessary to apply lower order control measures. The game reflects these constraints by implying some hazards require assistance and time to be resolved that is outside of the abilities of the player. As the player is learning about how to respond to hazards, the amount of helpful information can be structured to taper off as the game progresses in difficulty.

Feedback		
Question	Real Life	Game
F1: What is the result / new state of the environment once the choice and actions are made? [24]	A hazards risk will increase, decrease or remain the same.	The chance of an NPC being injured will increase, decrease or remain the same.
F2: How do I know I made the correct/incorrect choice?[21,24]	Feedback to convey a correct choice is often implicit. Incorrect choices are often explicitly conveyed through injury reports or workplace announcements.	Audio-visual effects, and removal of the hazard will convey that the player made correct choices. Injured NPCs and end game scores will convey incorrect choices.
F3: How will this affect my choices in the future? [24]	Without clear, explicit feedback or relevant prior knowledge, my future choices will likely remain unchanged.	Immediate and delayed explicit feedback will encourage the player to makes correct choices when presented with similar situations.

Table 5. Feedback Mapping Table

F1: The events that occur after a control measure is applied in the game will update the chances of an injury occurring. The aim is to encourage the player to re-engage in a new cycle of interaction with either the same or similar hazards.

F2: Feedback that a hazard is responded to correctly is limited in real life situation. The benefit to using a game-based learning approach is correct choices can be immediately re-enforced with feedback. Negative feedback in the game has been delayed i.e. an NPC injures themselves to show the real implications of failing to apply

effective control measures. Auxiliary audio and visual feedback mechanisms are also used to show that the user has performed an action the system recognises. It is expected that the player's motivations to improve the safety of the virtual environment will transfer to real motivations to improve the safety real workplaces they enter.

F3: In real life contexts, it is rare that an individual would have faced and responded to many different hazards in the workplace. Games can allow players to learn from their actions in a safe manner. The design of the game ensures the player regularly interacts with the core-gameplay cycle consisting of identifying and responding to hazards. This design decision ensures the resulting explicit feedback is received more readily and built upon so that future choices in the real world may be influenced by the experience.

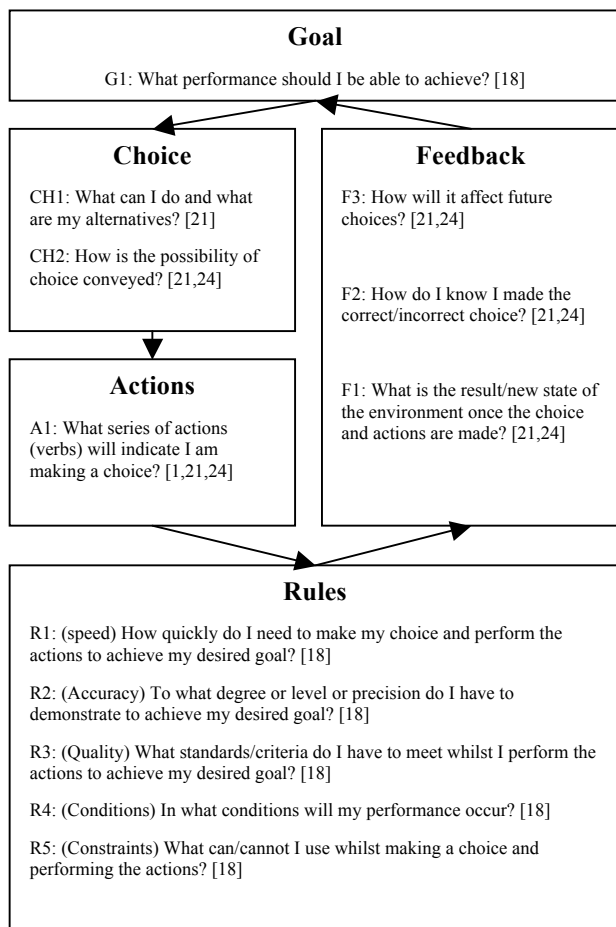


Figure 2. Core-Gameplay Framework with Design Questions

DISCUSSION AND CONCLUSION

The framework presented in this paper provides a clear process to map instructional objectives to core-gameplay activity. This design framework can work alongside existing frameworks and allows designers to systematically design core-gameplay that embeds educational objectives by using an inquiry design process. This inquiry design

process asks questions relating to the core-gameplay elements of goals, choices, actions, rules and feedback in relation to the instructional objectives (see Figure 2). As demonstrated in the example mapping tables, the answers to these questions form design criteria that allow designers to justify the learning potential of the game.

A limitation of this research is that the game example used has not been empirically tested to determine the extent to which it meets the identified learning objectives. Future research will aim to provide this empirical evidence. Additionally, there has been limited use of the framework in other case examples. In the future the framework could be applied to not just a training purpose, but also to the purposes of game-based awareness and assessment [10].

It is planned to apply a similar approach adopted by Arnab et al., [3] who repurposed their framework to focus on the analysis of serious games. The framework can be used as a flexible plug-in to add clarity to the design and analysis process articulated in existing frameworks such as LM-GM [3] or the Game Object Model 2 [2]. Therefore, future research may focus on the ease with which the framework can be used in conjunction with existing frameworks. Feedback from observing how others come to use the framework will only strengthen and enhance it to ensure it can significantly improve the way educational games are designed.

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